

Basic Principles

Before you proceed to study the trade functions of the platform, you must have a clear understanding of the basic terms: order, deal and position.

- An order is an instruction given to a broker to buy or sell a financial instrument. There are two main **types of orders**: Market and Pending. In addition, there are special **Take Profit** and **Stop Loss** levels.
- A deal is the commercial exchange (buying or selling) of a financial security. Buying is executed at the demand price (Ask), and Sell is performed at the supply price (Bid). A deal can be opened as a result of market order execution or pending order triggering. Note that in some cases, execution of an order can result in several deals.
- A position is a trade obligation, i.e. the number of bought or sold contracts of a financial instrument. A long position is financial security bought expecting the security price go higher. A short position is an obligation to supply a security expecting the price will fall in future.

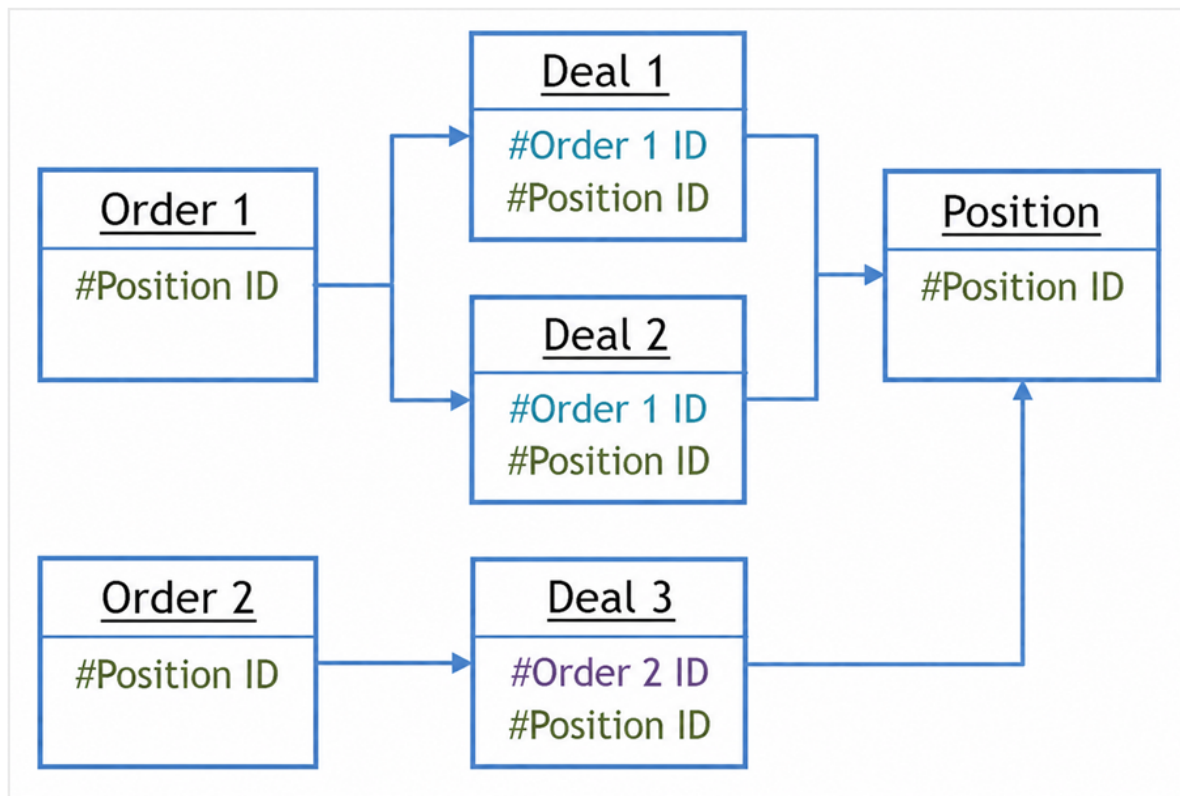
Interrelation of orders, deals and positions

The platform allows you to easily track how a position was opened or how a deal was performed. Each trading operation has its unique ID called a "ticket". Each order and deal receive a ticket relating to their relevant position. Each deal receives a ticket of an order, by which it was concluded.

If a position was affected by multiple deals, for example in the case of a partial closing or increasing volumes, each of the deals feature the position's ticket.

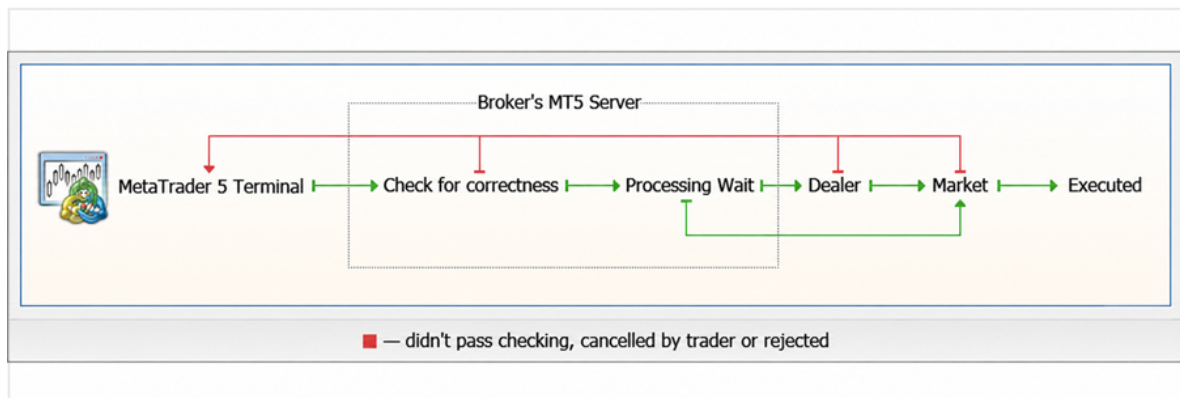
This makes it easy to track the entire history of the position as a whole.

If trading operations are sent to an exchange or a liquidity provider, they additionally feature an ID from an external system. This allows additional tracking of the interrelation of operations away from the platform.



A General Scheme of Trading Operations

- From the trading platform, an order is sent to a broker to execute a deal with the specified parameters;
- The correctness of an order is checked on the server (correctness of prices, availability of funds on the account, etc.);
- Orders that have passed the check wait to be processed on the trade server. Then the order can be:
 - executed (in one of automatic **execution** modes or by a dealer)
 - canceled upon expiry
 - rejected (e.g. when money is not enough or there is no suiting offer in the market; or rejected by the dealer)
 - canceled by a trader;
- A deal is the result of the execution of a market order or triggering of a pending order;
- If there are no positions for a symbol, conclusion of a deal results in opening of a position. If there is a position for the symbol, the deal can increase or reduce the position volume, close the position or reverse it.



Position Accounting System

Two position accounting systems are supported in the trading platform: Netting and Hedging. The system used depends on the account and is set by the broker.

Netting System

With this system, you can have only one common position for a symbol at the same time:

- If there is an open position for a symbol, executing a deal in the same direction increases the volume of this position.
- If a deal is executed in the opposite direction, the volume of the existing position can be decreased, the position can be closed (when the deal volume is equal to the position volume) or reversed (if the volume of the opposite deal is greater than the current position).

It does not matter, what has caused the opposite deal — an executed market order or a triggered pending order.

The below example shows execution of two EURUSD Buy deal 0.5 lots each:

Toolbox										×
Symbol	Ticket	Time	Type	Volume	Price	S / L	T / P	Price	Profit	
eurusd	8599748903	2026.05.12 18:58:06	buy	1	1.17332	1.16329 ×	1.18329 ×	1.17340	8.00	×
Balance: 9 516.39 USD Equity: 9 524.39 Margin: 1 173.32 Free Margin: 8 351.07 Margin Level: 811.75 %										8.00
Trade Exposure History News Mailbox Calendar Company Alerts Articles Code Base Experts Journal										

Execution of both deals resulted in one common position of 1 lot.

Hedging System

With this system, you can have multiple open positions of one and the same symbol, including opposite positions.

If you have an open position for a symbol, and execute a new deal (or a pending order triggers), a new position is additionally opened. Your current position does not change.

The below example shows execution of two EURUSD Buy deal 0.5 lots each:

Toolbox										×
Symbol	Ticket	Time	Type	Volume	Price	S / L	T / P	Price	Profit	
eurusd	8599537491	2026.05.12 18:45:59	buy	0.5	1.17325	1.16322 ×	1.17832 ×	1.17327	1.00	×
eurusd	8599543378	2026.05.12 18:46:17	buy	0.5	1.17318	1.16310 ×	1.18311 ×	1.17327	4.50	×
Balance: 9 514.89 USD Equity: 9 520.39 Margin: 1 173.22 Free Margin: 8 347.17 Margin Level: 811.48 %										5.50
Trade Exposure History News Mailbox Calendar Company Alerts Articles Code Base Experts Journal										

Execution of these deals resulted in opening two separate positions.

Impact of the System Selected

Depending on the position accounting system, some of the platform functions may have different behavior:

- **Stop Loss and Take Profit inheritance** rules change.
- To **close a position** in the netting system, you should perform an opposite trading operation for the same symbol and the same volume. To close a position in the hedging system, explicitly select the "Close Position" command in the context menu of the position.
- A position cannot be reversed in the hedging system. In this case, the current position is closed and a new one with the remaining volume is opened.
- In the hedging system, a new condition for margin calculation is available — **Hedged margin**.

Types of Orders

The trading platform allows to prepare and issue requests for the broker to execute trading operations. In addition, the platform allows to control and manage open positions. Several types of trading orders are used for these purposes. An order is a trader's instruction to the broker to perform a trade operation. In the platform, orders are divided into two main types: market and pending. In addition, there are special **Stop Loss** and **Take Profit** orders.

Market Order

A market order is an instruction given to a brokerage company to buy or sell a financial instrument. Execution of this order results in the execution of a deal. The price at which the deal is executed is determined by the **type of execution** that depends on the symbol type. Generally, a security is bought at the Ask price and sold at the Bid price.

Pending Order

A pending order is the trader's instruction to a brokerage company to buy or sell a security in future under pre-defined conditions. The following types of pending orders are available:

- **Buy Limit** — a trade request to buy at the Ask price that is equal to or less than that specified in the order. The current price level is higher than the value specified in the order. Usually this order is placed in anticipation of that the security price will fall to a certain level and then will increase;
- **Buy Stop** — a trade order to buy at the "Ask" price equal to or greater than the one specified in the order. The current price level is lower than the value specified in the order. Usually this order is placed in the anticipation that the price will reach a certain level and will continue to grow;
- **Sell Limit** — a trade order to sell at the "Bid" price equal to or greater than the one specified in the order. The current price level is lower than the value specified in the order. Usually this order is placed in anticipation of that the security price will increase to a certain level and will fall then;

- **Sell Stop** — a trade order to sell at the "Bid" price equal to or less than the one specified in the order. The current price level is higher than the value in the order. Usually this order is placed in anticipation of that the security price will reach a certain level and will keep on falling.
- **Buy Stop Limit** — this type is the combination of the first two types, being a stop order to place a Buy Limit order. As soon as the future Ask price reaches the stop-level indicated in the order (the Price field), a Buy Limit order will be placed at the level, specified in Stop Limit price field. A stop level is set above the current Ask price, while Stop Limit price is set below the stop level.
- **Sell Stop Limit** — this order is a stop order to place a Sell Limit order. As soon as the future Bid price reaches the stop-level indicated in the order (the Price field), a Sell Limit order will be placed at the level, specified in Stop Limit price field. A stop level is set below the current Bid price, while Stop Limit price is set above the stop level.

For symbols with Exchange Stocks, Exchange Futures and Futures Forwards **calculation modes**, all types of pending orders are triggered according to the rules of the exchange where trading is performed. Usually, Last price (price of the last performed transaction) is applied. In other words, an order triggers when the Last price touches the price specified in the order. But note that buying or selling as a result of triggering of an order is always performed by the Ask and Bid prices respectively.

In the "Exchange execution" mode, the price specified when placing limit orders is not verified. It can be specified above the current Ask price (for the Buy Limit orders) and below the current Bid price (for the Sell Limit orders). When placing an order with such a price, it triggers almost immediately and turns into a market one. However, unlike market orders where a trader agrees to perform a deal by a non-specified current market price, a pending order will be executed at a price no worse than the one specified.

If during pending order activation the corresponding market operation cannot be executed (for example, the free margin on the account is not enough), the pending order will be canceled and moved to **history** with the "Rejected" status.



1

— current market state

1

— forecast

●

— current price

●

— order price

■

— price, reaching which a pending order will be placed

↑

— expected growth

↓

— expected fall

Take Profit

The Take Profit order is intended for gaining the profit when the security price reaches a certain level. Execution of this order results in the complete closing of the entire position. It is always connected to an open position or a pending order. The order can be requested only together with a market or a pending order. This order condition for long positions is checked using the Bid price (the order is always set above the current Bid price), and the Ask price is used for short positions (the order is always set below the current Ask price).

Stop Loss

This order is used for minimizing losses if the security price moves the wrong direction. If the security price reaches this level, the entire position is closed

automatically. Such orders are always associated with an open position or a pending order. They can be requested only together with a market or a pending order. This order condition for long positions is checked using the Bid price (the order is always set below the current Bid price), and the Ask price is used for short positions (the order is always set above the current Ask price).

If during Take Profit or Stop Loss activation the corresponding market operation cannot be executed (for example, it is rejected by the exchange), the order will not be deleted. It will trigger again at the next tick corresponding to the order activation conditions.

Rules of Stop Loss and Take Profit inheritance (netting): #

- When a position volume is increased or the position is reversed, Take Profit and Stop Loss are placed according to its latest order (market or triggered pending order). In other words, stop levels in each subsequent order of the same position replace previous ones. If zero values are specified in the order, Stop Loss and Take Profit of a position will be deleted.
- If a position is partially closed, Stop Loss and Take Profit are not changed by the new order.
- If a position is fully closed, the Stop Loss and Take Profit levels are deleted, because they are associated with an open position and cannot exist without it.
- When a trade operation is executed for a symbol, for which there is a position, the current Stop Loss and Take Profit of the open position are automatically inserted in the order placing window. This is aimed to prevent accidental deletion of current stop orders.
- During one click trading operation (from a **panel on the chart** or from the **Market Watch**) for the symbol, for which there is a position, the current values of Stop Loss and Take Profit are not changed.
- On the OTC markets (Forex, Futures), when a position is moved to the next trading day (the swap), including swap through re-opening, the levels of Stop Loss and Take Profit remain unchanged.
- On the exchange market, when a position is moved to the next trading day (the swap), as well as when moved to another account or during delivery, the levels of Stop Loss and Take Profit are reset.

Stop Loss and Take Profit inheritance rule (hedging):

- If a position is partially closed, Stop Loss and Take Profit are not changed by the new order.
- If a position is fully closed, the Stop Loss and Take Profit levels are deleted, because they are associated with an open position and cannot exist without it.
- During one click trading operation (from a **panel on the chart** or Depth of Market), the Stop Loss and Take Profit levels are not set.

These rules apply both when trading manually and when placing orders from **Expert Advisors** (MQL5 programs).

Trailing Stop can be used to make Stop Loss follow the price automatically.

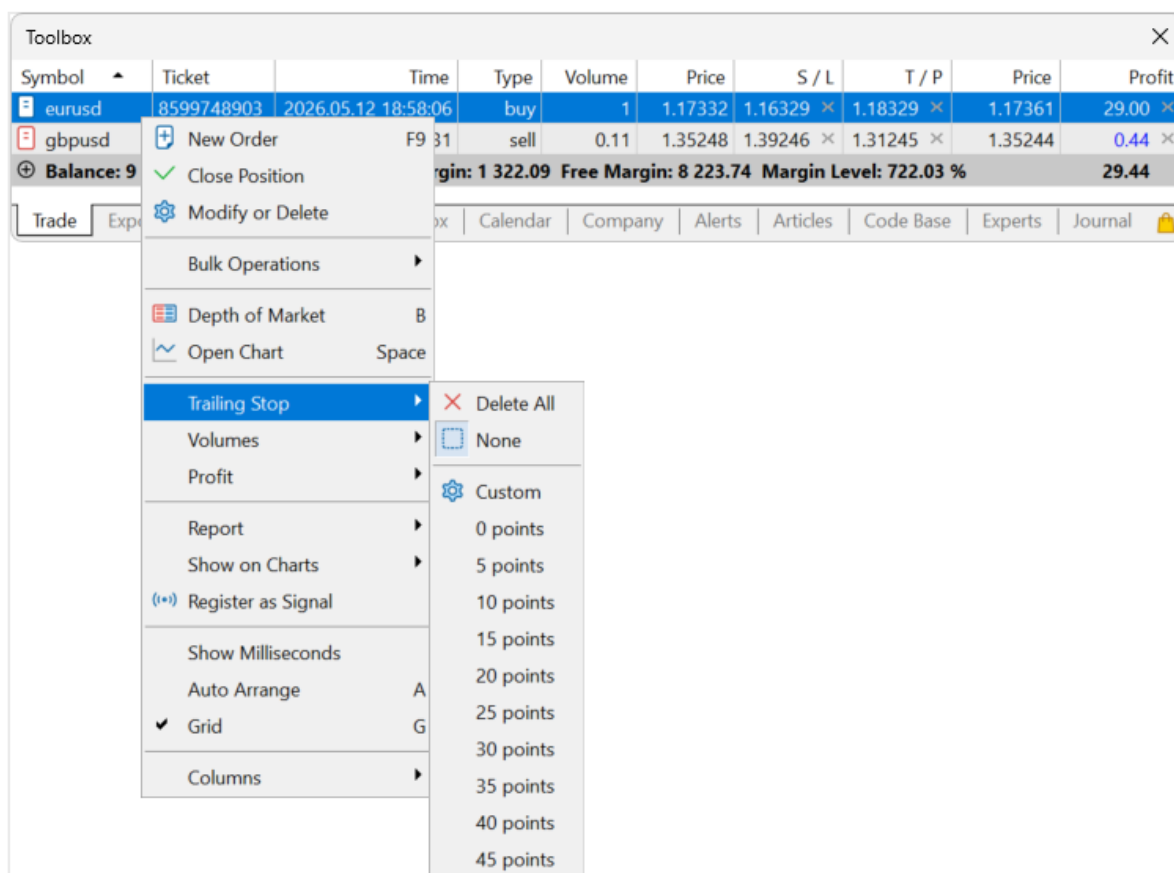
Activation of Take Profit or Stop Loss results in the complete closing of the entire position.

For symbols with Exchange Stocks, Exchange Futures and Futures Forwards **calculation modes**, Stop Loss and Take Profit orders are triggered according to the rules of the exchange where trading is performed. Usually, Last price (price of the last performed transaction) is applied. In other words, a stop-order triggers when the Last price touches the specified price. However note that buying or selling as a result of activation of a stop-order is always performed by the Bid and Ask prices.

Trailing Stop

Stop Loss is used for minimizing losses if the security price moves the wrong direction. Once a position becomes profitable, its Stop Loss can be manually moved to a break-even level. Trailing Stop automates this process. This tool is especially useful during a strong unidirectional price movement or when it is impossible to monitor the market continuously for some reason.

Trailing Stop is always associated with an **open position** or a **pending order**. It is executed in the trading platform rather than on the server like Stop Loss. To set a Trailing Stop, select "Trailing Stop" in the context menu of a position or an order in the **"Trading"** tab:



Select a necessary value of a distance between the Stop Loss level and the current price.

For each open position or pending order only one Trailing Stop can be set.

Scheme of Trailing Stop Operation

- When new quotes arrive, the platform checks whether an open position is profitable.
- As soon as the profit in points becomes equal to or larger than the indicated level, an automatic command is generated to place a Stop Loss at the indicated distance from the current price.
- If price moves increasing the position profit, "Stop Loss" automatically moves together with the price.
- Otherwise, the order is not modified. Thus, the profit of a trading position is fixed automatically.
- If a Stop Loss has been set for the position, it also follows the price when the position profit increases and remains unchanged if it decreases.

When a pending order triggers, the trailing stop of the current position for the same symbol is overwritten with the trailing stop specified for the order.

If a deal made as a result of triggering of a pending order has the opposite direction to the current position for the symbol and has less or equal volume, then the trailing stop is not overwritten.

With each automatic modification of Stop Loss an entry is added to the journal. To disable Trailing Stop, set the "



None" parameter in the control menu. The "



Delete All" command disables Trailing Stops of all open positions and **pending orders**.

The Trailing Stop is executed in the trading platform rather than on the server (like Stop Loss or Take Profit). This is why it will not work, unlike the above orders, if the platform is off. In this case, only the Stop Loss level set by the Trailing Stop will trigger.

For one position, Trailing Stop cannot occur more than once every 10 seconds.

If there are several positions with Trailing Stop at a single symbol, the Trailing Stop is processed in a specific way. When a tick arrives, only a Trailing Stop of the last opened position is processed. If yet another tick arrives for the same symbol within 10 seconds, a Trailing Stop of the next position (opened second last) is processed. If the next tick arrives later than 10 seconds, a Trailing Stop of the position opened last is processed again.

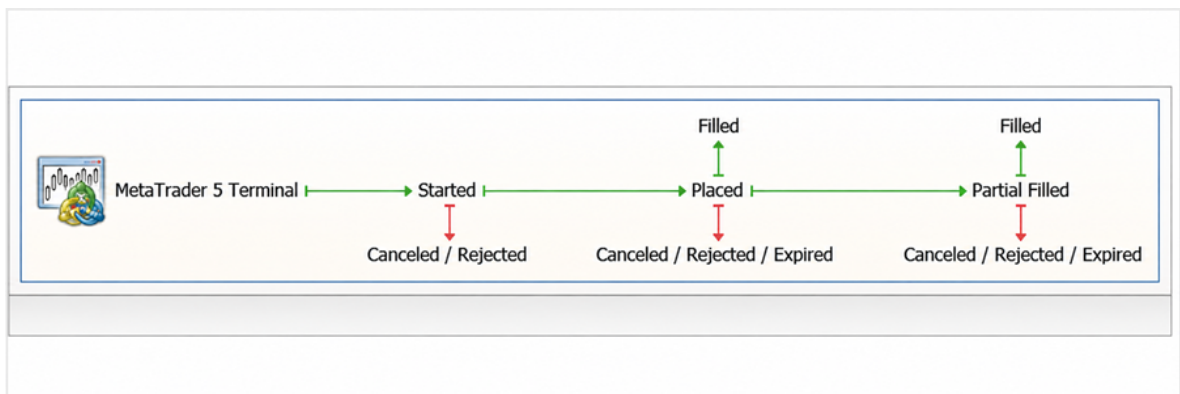
State of Orders #

After an order has been formed and sent to a trade server, it can undergo the

following stages:

- Started — the order correctness has been checked, but it hasn't been yet accepted by the broker;
- Placed — a dealer has accepted the order;
- Partially filled — the order is filled partially;
- Filled — the entire order is filled;
- Canceled — the order is canceled by the client;
- Rejected — the order is rejected by a dealer;
- Expired — the order is canceled due to its expiration.

You can view the state of orders on the "History" tab in field "State". The state of pending orders that haven't triggered yet can be viewed on the "Trade" tab.



Types of Execution

Four **order** execution modes are available in the trading platform:

- **Instant Execution**

In this mode, an order is executed at the price offered to a broker. When sending an order to be executed, the platform automatically adds the current prices to the order. If the broker accepts the prices, the order is executed. If the broker does not accept the requested price, a "Requote" is sent — the broker returns prices, at which this order can be executed.

- **Request Execution**

In this mode, a market order is executed at the price previously received from a broker. Prices for a certain market order are requested from the broker before the order is sent. After the prices have been received, order execution at the given price can be either confirmed or rejected.

- **Market Execution**

In this order execution mode, a broker makes a decision about the order execution price without any additional discussion with a trader. Sending an order in such a mode means advance consent to its execution at this price.

- **Exchange Execution**

In this mode, trade operations conducted in the trading platform are sent to an external trading system (exchange). Trade operations are

executed at the prices of current market offers.

Execution mode for each security is defined by the brokerage company.

Fill Policy

In addition to common rules of order execution set by a broker, a trader can indicate additional conditions in the "Fill Policy" field of the order placing window:

- Fill or Kill (FOK)

This fill policy means that an order can be filled only in the specified volume. If the necessary amount of a financial instrument is currently unavailable in the market, the order will not be executed. The required volume can be filled by several offers available in the market at the moment.

- Immediate or Cancel (IOC)

In this case a trader agrees to execute a deal with the volume maximally available in the market within that indicated in the order. In case the order cannot be filled completely, the available volume of the order will be filled, and the remaining volume will be canceled. The possibility of using IOC orders is determined on the trade server.

- Book or Cancel (BOC)

The BOC policy indicates that the order can only be placed in the Depth of Market (order book). If the order can be filled immediately when placed, this order is canceled. This policy guarantees that the price of the placed order will be worse than the current market. BOC is used to implement passive trading: it is guaranteed that the order cannot be executed immediately when placed and thus it does not affect current liquidity. This fill policy is only supported for limit and stop limit orders.

- Return

This policy is only used for market (Buy and Sell), limit and stop limit orders. If filled partially, an order with the remaining volume is not canceled, and is processed further. For market orders, the Return policy is used only in the Exchange Execution mode, while for limit and stop limit ones, it is applied in the Market Execution and Exchange Execution modes.

Use of fill policies depending on the execution type can be shown as the following table:

Type of Execution/ Fill Policy	Fill or Kill	Immediate or Cancel	Book or Cancel	Return
Instant Execution	+	—	—	—
Request Execution	+	—	—	—

Market Execution	+	+	—	+
Exchange Execution	+	+	+	+

Trading Report

The report represents trading results in a convenient visual format. It assists in evaluating trading performance, optimizing portfolios and finding methods to achieve lower risks along with improved trading stability. To analyze your strategy, click 'Report \ Overview' in the context menu of the trading history section or Reports in the View menu.

The report is divided into tabs, each providing aggregated information:

- **Summary** — an overview of trading over time: overall profit and loss metrics, deposit and withdrawal amounts, balance, growth and dividends graphs, and other trading results.
- **Profit/Lost** — details on profitable and losing trades, categorized by trading types (manual, algorithmic and copy trading). Results can be analyzed in the context of trades or money by months and years.
- **Long/Short** — dynamic ratio of Buy and Sell trades at specified time periods, as well as profitability of Buy and Sell directions.
- **Symbols** — analysis of trades by financial instruments, showing which symbols yield gains or losses, how often you trade them, and graphs of trades and monetary volumes.
- **Risks** — key risk characteristics of your strategy: drawdown and deposit load graphs, and the ratio of profitable and losing trades.

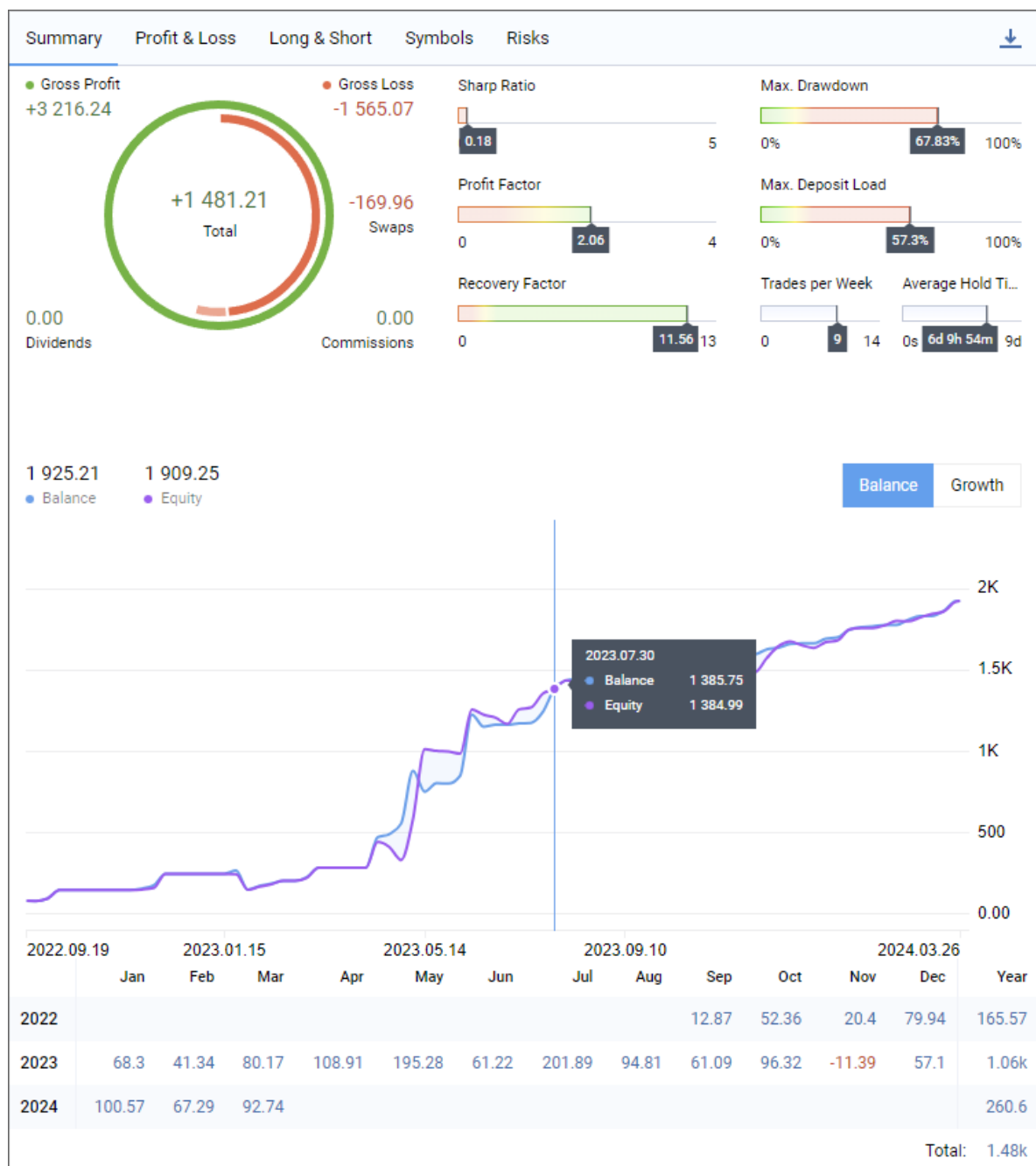
For recommendations on how to use the report refer to the article "[New MetaTrader report: 5 most important trading metrics](#)".

The report is generated based on the [trading history](#) of the current account. To analyze trading activity on a different account, switch to it and then reopen the report.

Some metrics in the report require historical price data of financial instruments. For example, to display graphs of funds, deposit load and drawdowns, the platform calculates the fluctuations in floating profits across positions over time. Consequently, when you open a report for the first time, the platform may require some time to load prices and perform calculations. Next time the report will open instantly.

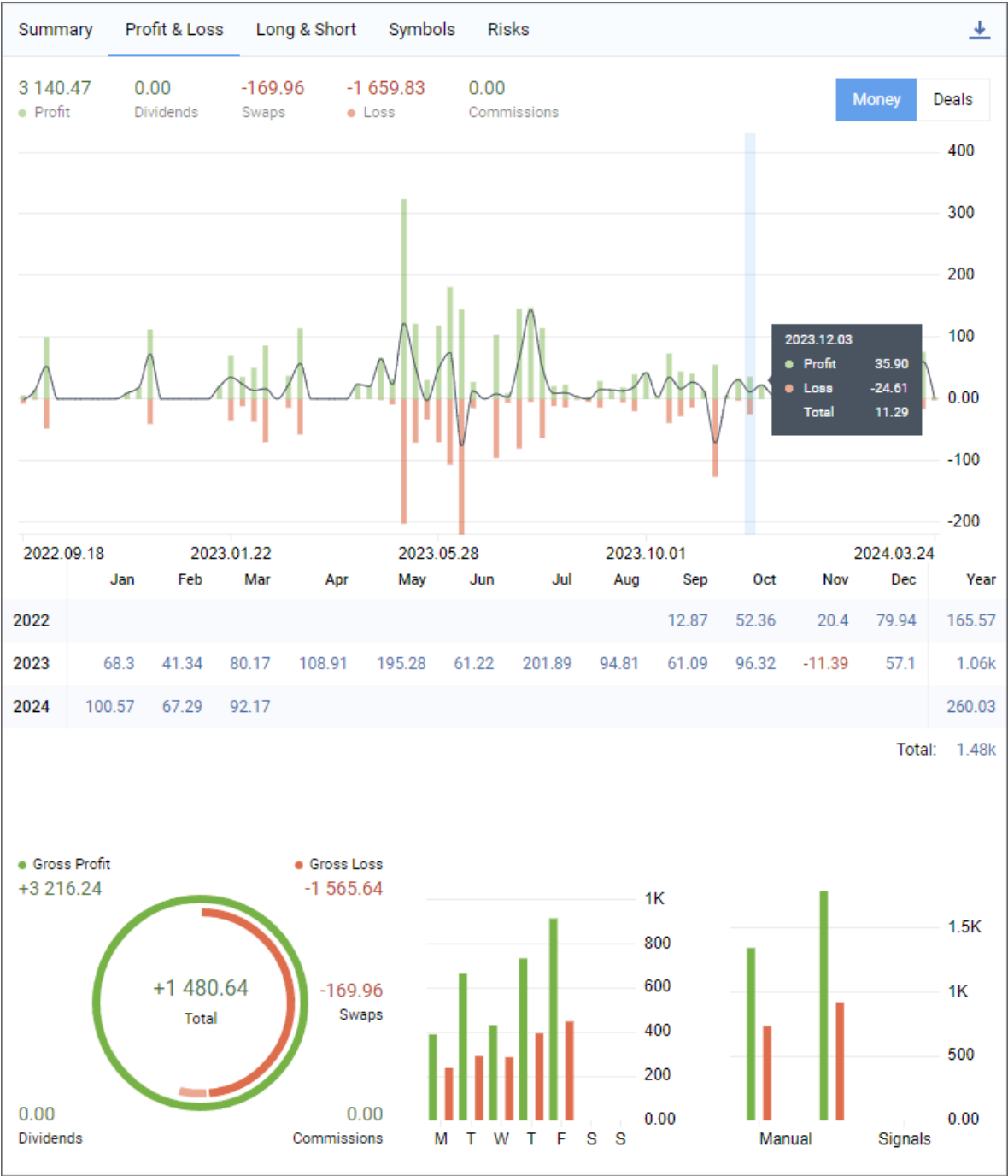
Summary

This section provides an overview of trading over time: overall profit and loss metrics, deposit and withdrawal amounts, balance, growth and dividends graphs, and other trading results.



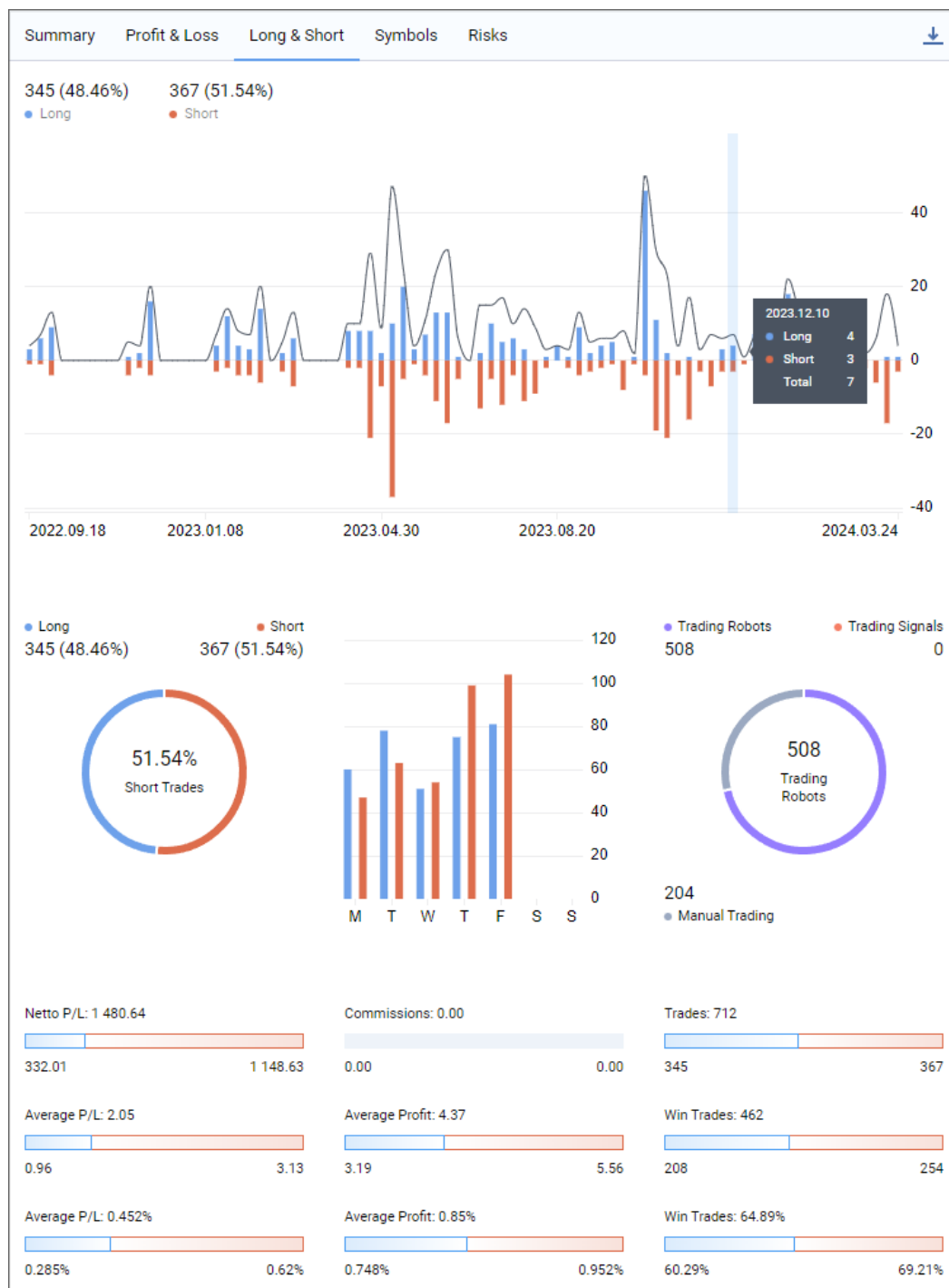
Profit/Loss

This section displays information about profitable and losing trades, categorized by trading types (manual, algorithmic and copy trading). Results can be analyzed in the context of deals or money by month and year.



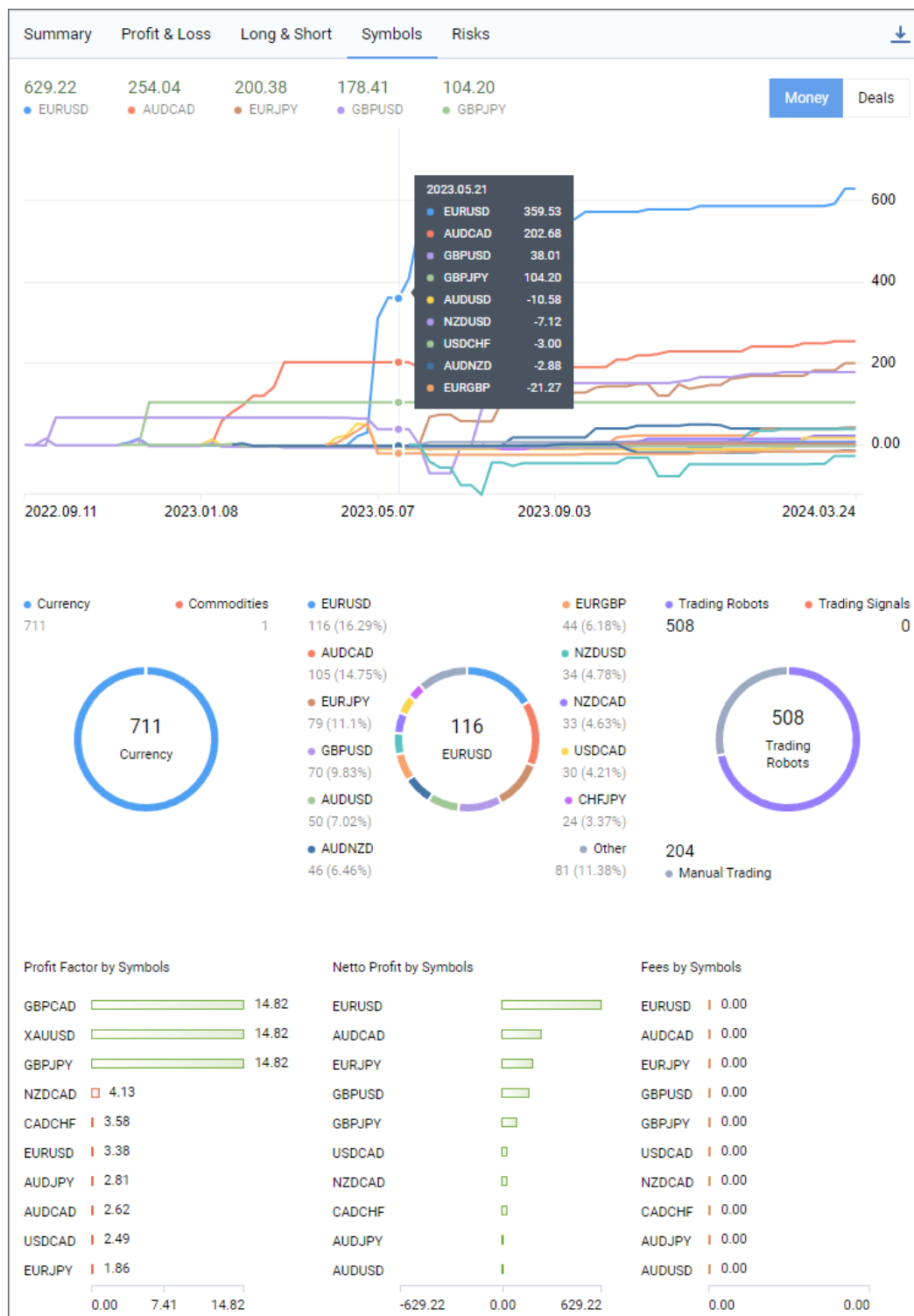
Long/Short #

This section shows the dynamic ratio of Buy and Sell trades at specified time periods, as well as profitability of Buy and Sell directions.



Symbols

This section enables the analysis of trades in terms of financial instruments. showing which symbols yield gains or losses, how often you trade them, and graphs of trades and monetary volumes.



Risks

This section visualizes key risk characteristics of your strategy: drawdown and deposit load graphs, and the ratio of profitable and losing trades.

Drawdown/Deposit Load graphs

Depending on the selected mode, the section displays:

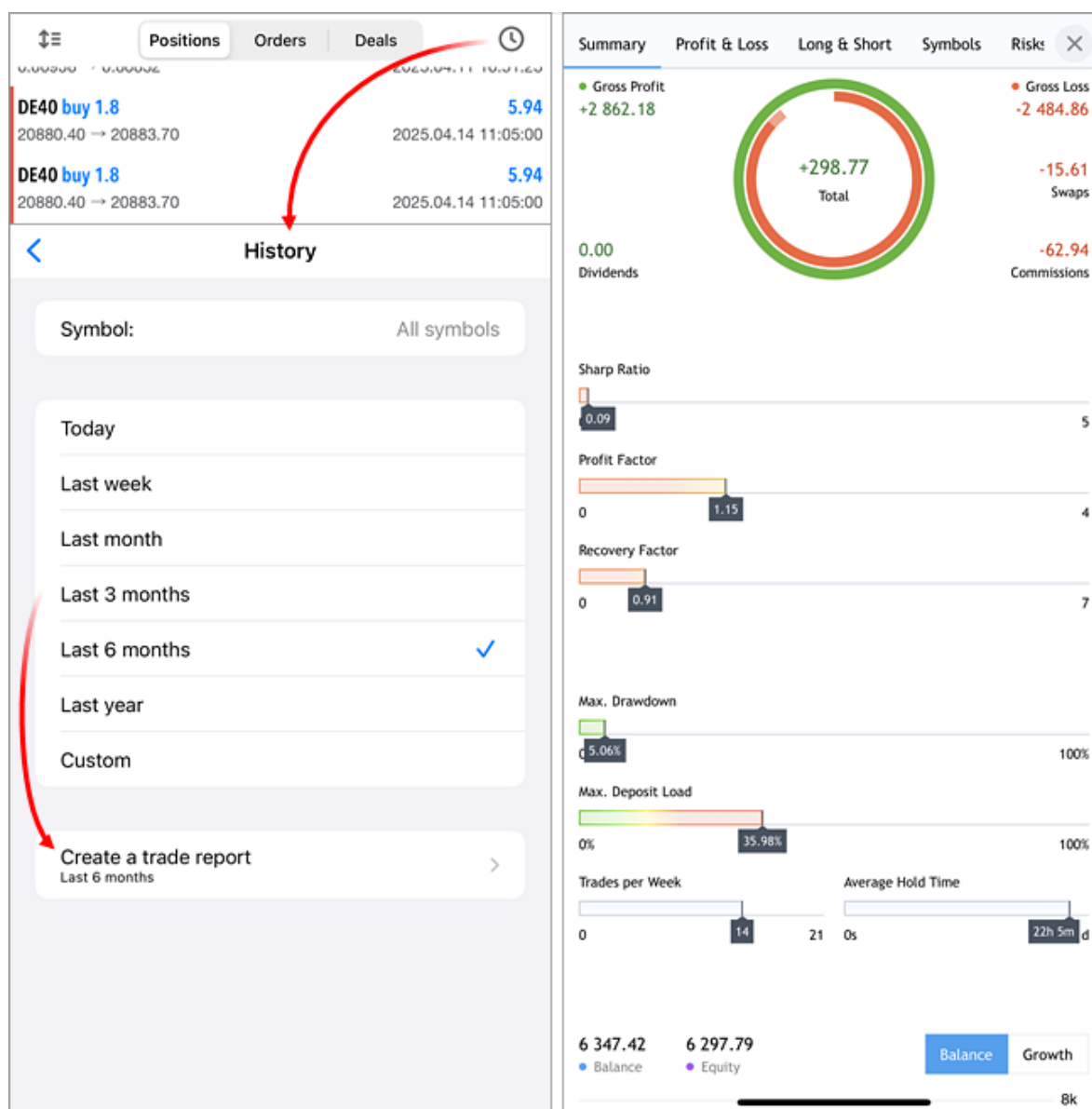
- Drawdown graph applied over the balance change graph.
- Deposit Load graph applied over the balance change graph.

By analyzing these metrics, you can evaluate how aggressive your trading is and whether this style is justified.



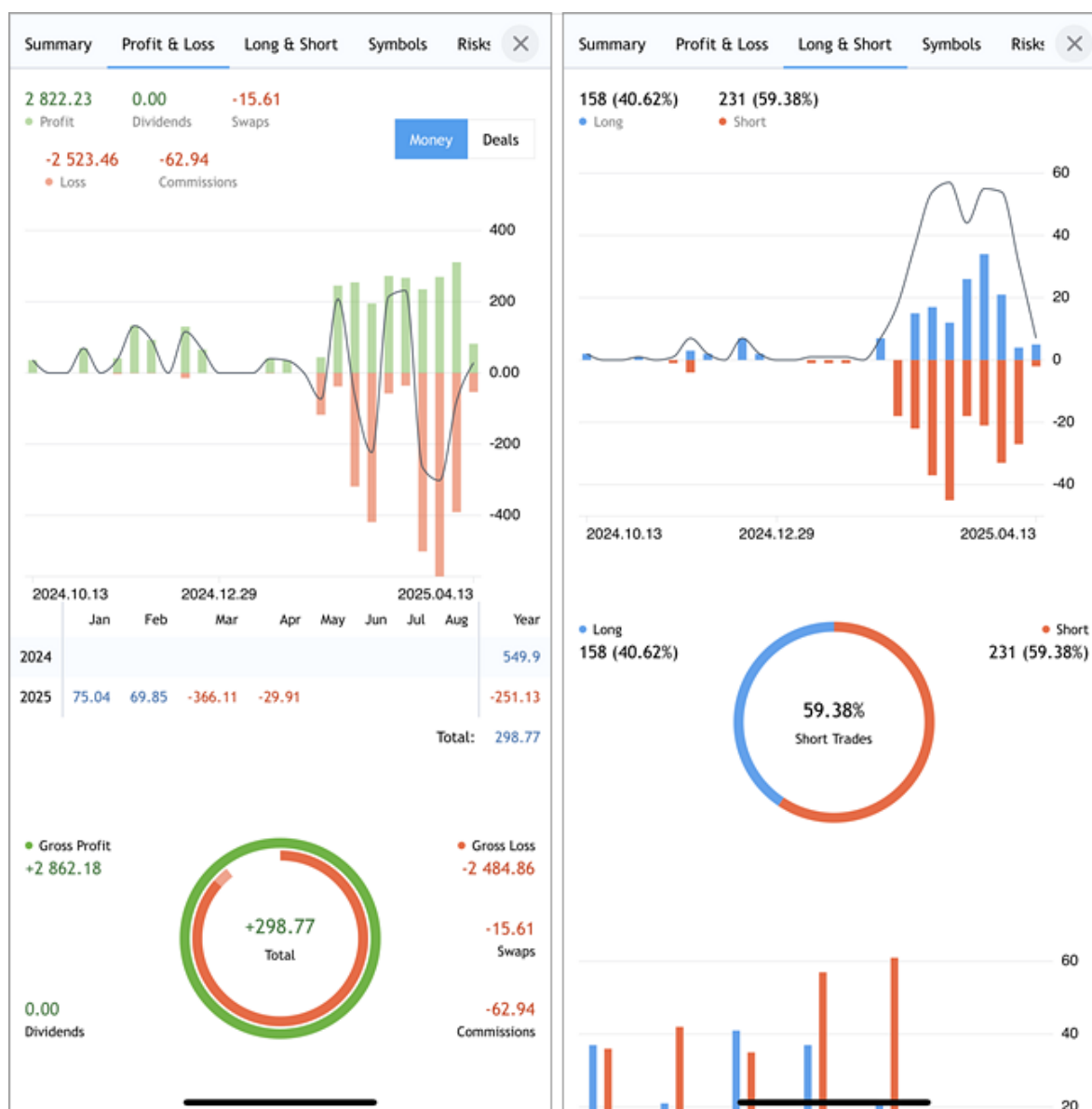
Trading Report

The report represents trading results in a convenient visual format. It assists in evaluating trading performance, optimizing portfolios and finding methods to achieve lower risks along with improved trading stability. To analyze your strategy, go to the history section and click on the period selection icon. Next, select the period and click "Create a trade report".



The report is divided into tabs, each providing aggregated information:

- Summary – an overview of trading over time: overall profit and loss metrics, deposit and withdrawal amounts, balance, growth and dividends graphs, and other trading results.
- Profit/Lost – data on profitable and losing trades. The parameter is divided by types of trading (manual, algorithmic and copying trades). The results can be analyzed in terms of trades or money by months and years.
- Long/Short – dynamic ratio of buy and sell operations at specified periods of time, as well as Buy and Sell profitability.
- Symbols – analysis of trades by financial instruments. You will see here which symbols you earn or lose on, how often you trade them, graphs of trades and monetary volumes for them.
- Risks – key risk characteristics of your strategy: drawdown and deposit load graphs, and the ratio of profitable and losing trades.



To save the report file or share it via messenger, email, or social media, scroll through all the report tabs to the end and tap



. Next, select "Save as PDF". This will open your device's standard file handling menu.

For recommendations on how to use the report, refer to the article "[New MetaTrader report: 5 most important trading metrics](#)". A full description of all report parameters is available in the [desktop platform's Help](#) section.

The report is generated based on the **trading history** of the current account. To analyze trading activity on a different account, switch to that account first, then reopen the report.

To calculate certain metrics, the platform uses historical price data for financial instruments. For example, to display equity, deposit load, and drawdown graphs, it calculates how floating profit across positions has changed over time. Consequently, the first time you open the report, the platform may require some time to download prices and perform calculations. Next time the report will open instantly.

The price data used for generating the report may consume a significant amount of your device's memory. If you encounter memory limitations, try generating the report for a shorter time period.

History

The history of trade operations performed on a trading account is available on this tab. Use the bottom panel of the application to switch to this tab.

There are three operation viewing modes: Orders, Deals and Positions. You can switch between them at the top of the tab.

Orders

By default, only brief information about orders is shown here.

The following summary information is displayed at the bottom of the screen: the total number of orders, and the number of **filled and canceled** orders.

Positions Orders Deals		
EURUSD, buy	2017.06.26 16:54:03	
0.57 / 0.57 at 1.12104		filled
AUDUSD, sell stop	2017.06.26 16:54:15	
1.00 / 1.00 at 0.75973		filled
USDCHF, sell	2017.06.26 16:54:30	
0.57 / 0.57 at 0.96997		filled
EURUSD, sell	2017.06.30 15:36:18	
0.57 / 0.57 at 1.14079		filled
USDCHF, buy	2017.06.30 15:36:24	
0.57 / 0.57 at 0.95864		filled
AUDUSD, buy	2017.06.30 15:36:38	
1.00 / 1.00 at 0.76794		filled
EURUSD, buy	2017.06.30 15:36:56	
1.00 / 1.00 at 1.14115		filled
EURUSD, sell	2017.06.30 15:37:13	
1.00 / 1.00 at 1.14124		filled
EURUSD, buy	2017.06.30 15:37:32	
1.00 / 1.00 at 1.14131		filled
USDCHF, sell	2017.06.30 15:38:00	
1.00 / 1.00 at 0.95815		filled
Filled	10	
Canceled	0	
Total	10	
Quotes Chart Trade History Settings		

To see more details, tap once on the order line.

EURUSD sell		#50117259331
Euro vs US Dollar		
1 / 1 at 1.07345		filled
2022.05.24 10:44:42		
S/L: -		
T/P: -		
[tp 1.07345]		

1. Symbol – a financial instrument of the order.
2. Type – **type of the order**: "Buy", "Sell", "Sell Stop", "Sell Limit", "Buy Stop", "Buy Limit", "Buy Stop Limit" or "Sell Stop Limit".
3. # – the ticket number (a unique identifier) of a trade operation.
4. Volume – volume requested in the order (in lots or units). The minimal

volume and its change step are limited by a brokerage company, the maximal value is limited by the deposit size.

5. Price – price specified in the order at which the trade operation should be executed.
6. State – **result** of the order placing: "Filled", "Partially", "Canceled" etc.
7. Time – the time when the order was placed by the trader. The record is represented as YYYY.MM.DD HH:MM (year.month.day hour:minute).
8. S/L – level of the placed **Stop Loss order**. If a trade position was closed by Stop Loss, the order is colored in red. If the order was not placed, a dash is displayed.
9. T/P – level of the set **Take Profit** order. If a trade position was closed by Take Profit, the order is colored in green. If the order was not placed, a dash is displayed.
10. Comment – a comment to the order. A comment can only be added when **placing an order** from the PC version of the MetaTrader 5 platform. The comment cannot be changed when modifying an order. In addition, a comment on a trade operation can be added by a brokerage company. If an order was executed as a result of Stop Loss or Take Profit triggering, the following comment is added automatically: [sl price] or [tp price].

Positions	Orders	Deals	
EURUSD, buy, in		2017.06.26 16:54:03	
0.57 at 1.12104			
AUDUSD, sell, in		2017.06.26 16:54:15	
1.00 at 0.75973			
USDCHF, sell, in		2017.06.26 16:54:30	
0.57 at 0.96997			
EURUSD, sell, out		2017.06.30 15:36:18	
0.57 at 1.14079			1 125.75
USDCHF, buy, out		2017.06.30 15:36:24	
0.57 at 0.95864			673.67
AUDUSD, buy, out		2017.06.30 15:36:38	
1.00 at 0.76794			-821.00
EURUSD, buy, in		2017.06.30 15:36:56	
1.00 at 1.14115			
EURUSD, sell, out		2017.06.30 15:37:13	
1.00 at 1.14124			9.00
EURUSD, buy, in		2017.06.30 15:37:32	
1.00 at 1.14131			
USDCHF, sell, in		2017.06.30 15:38:00	
1.00 at 0.95815			
Deposit		1 000 000.00	
Profit		987.42	
Swap		-94.76	
Commission		0.00	
Balance		1 000 892.66	
			
Quotes	Chart	Trade	History
			 4
			Settings

Deals

By default, only brief information about deals is shown here.

If a deal was executed as a result of **Stop Loss** order activation, it is colored in red. If **Take Profit** triggered, the deal is colored in green. In the bottom line, the result of deal execution for the **selected period** is shown:

Deposit – the sum of deposits and withdrawals.

Profit – the total profit/loss obtained from deals for the selected period (swaps and commissions are not included).

Swap – the total amount of added/charged swaps.

Commission – the total amount of commissions taken for execution of deals.

Balance – the total balance by operations performed for the selected period.

To see more details, tap once on the deal line.

Symbol – a financial instrument of the deal.

Type – type of a trade operation: "Buy" – a buy deal, "Sell" – a sell deal, "Balance" – a balance operation.

Direction – direction of the deal relative to the current position on a particular symbol: "in", "out" or "in/out".

– ticket number (a unique identifier) of a deal.

Volume – volume of an executed deal (in lots or units).

Price – the deal was executed at this price.

Profit – the financial result of position exiting. For entry deals, zero profit is shown.

Δ – the difference between the position open price and its close price in points and in percentages. The value is only shown for market exit trades.

Time – time of the deal. The record is represented as YYYY.MM.DD HH:MM (year.month.day hour:minute).

Order – ticket number (a unique identifier) the trade was executed on. Several deals can correspond to one order, if the required volume specified in the order was not covered by one market offer.

Swap – the size of the swap.

Commission – commission charged for the deal execution.

Comment – a comment to the deal. A comment can only be added when **placing an order** from the PC version of the MetaTrader 5 platform. The comment cannot be changed when modifying an order. In addition, a comment on a trade operation can be added by a brokerage company. If a deal was executed after the triggering of Stop Loss or Take Profit, the following comment is added automatically: [sl price] or [tp price].

Fee – separate fee charged by the broker in addition to commission.

Positions #

The platform collects data on deals related to a position (position opening, additional volume, partial and full closure), and then combines the data into one record providing the following details:

Position opening and closing time determined by the first and last trade respectively

Position volume. If part of the position was closed, the record contains the closed volume and the source volume

The weighted average open and close prices of the position

The total financial result of deals related to the position (swaps and commissions are not included)

To display the trading history as positions, the platform uses information about deals executed during the requested period. A position is included into the report only if its opening date or closing date falls within the selected period. If a position was opened before the start of the selected time period and was closed after its end, this position will not be included in the report.

Positions closed by Stop Loss or Take Profit are marked with red and green vertical lines.

Quick commands

Swipe left on the operation line to open the quick commands menu:

- perform a trading operation for the selected symbol
- open the chart of the selected symbol

By dragging the menu further to the left, you can immediately switch to the symbol chart.

Customize the History Appearance #

You can configure the depth of history and the symbols of operations shown in the history. To go to these settings, tap .

This window contains the following settings:

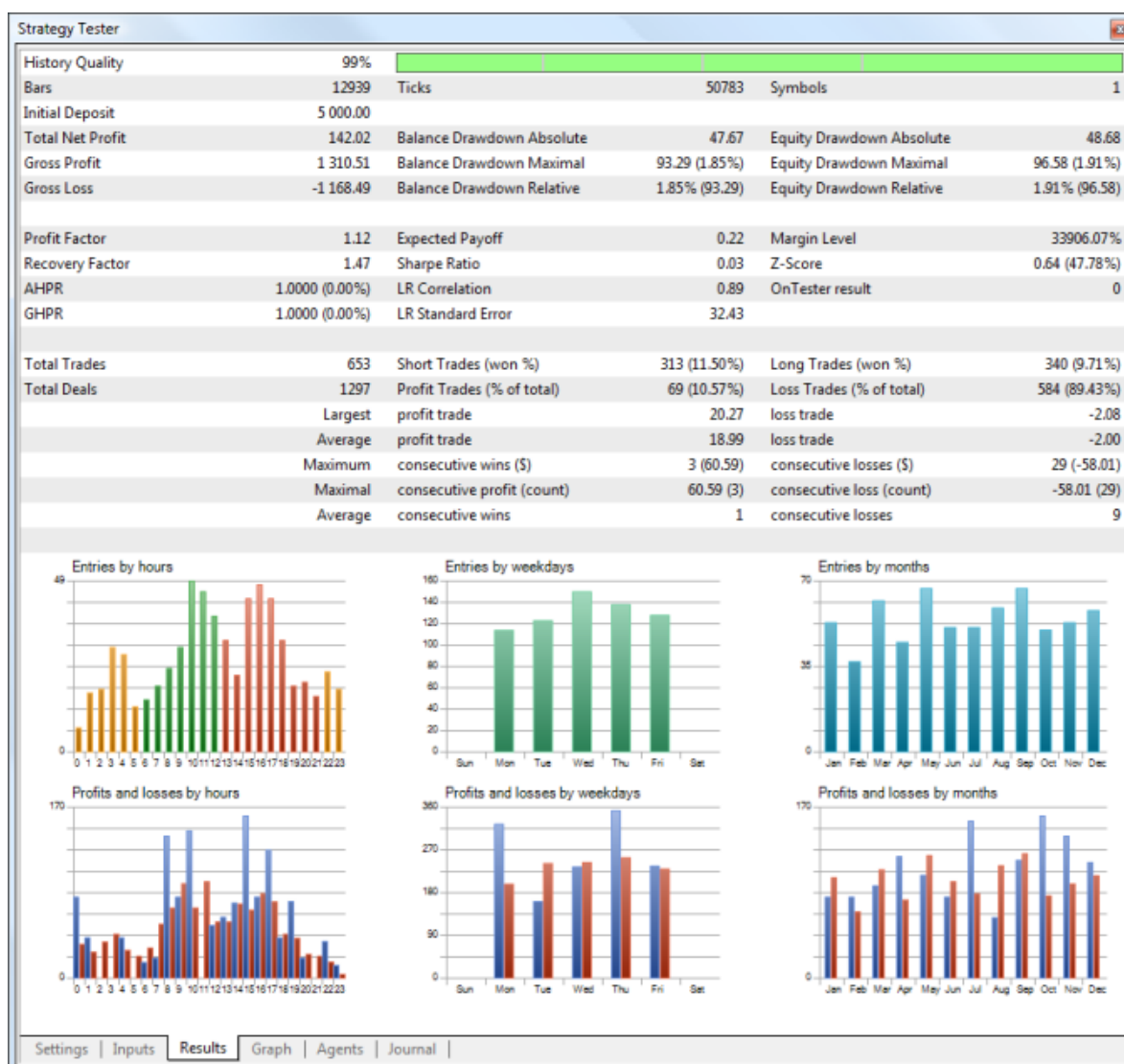
Symbol – select the symbol of trade operations to show in the history.

Period – the period, for which trade operations will be shown in the history. You can specify one of the predefined periods or enter a custom one. In the second case, the window will have additional fields "From" and "To", where you need to specify your time frames.

When you are finished, tap "Done".

Testing Report

You can view a detailed report on the "Results" tab.



The following parameters are available in the testing report:

- History Quality — this value characterizes the quality of price data used for testing. It is determined as a percentage ratio of correct and incorrect one-minute data. Bars with a volume equal to 1 with different OHLC values are considered incorrect. History gaps are also considered as incorrect data. Depending on size, the period of testing is divided into 1 — 199 intervals. The history quality is determined for each of them separately. The time intervals are shown in different colors on the graphical indicator of the history quality (the lighter tint of green means the better quality, the red color represents intervals with the quality lower than 50%).
- Bars — the number of bars generated for the testing **symbol**;
- Ticks — the number of ticks modeled during testing;
- Symbols — the number of symbols, for which information was requested by the Expert Advisor during testing;
- Initial Deposit — **initial deposit** for testing;
- Withdrawal — the amount of money withdrawn by an Expert Advisor during testing. This field is not displayed if there are no withdrawal operations;

- Total Net profit — the financial result of all trades.
- Gross Profit — the sum of all profitable trades in terms of money;
- Gross Loss — the sum of all losing trades in terms of money;
- Balance Drawdown Absolute — difference between the initial deposit and the minimal level below initial deposit throughout the whole testing period. $\text{AbsoluteDrawDown} = \text{InitialDeposit} - \text{MinimalBalance}$
See the [drawdown calculation example](#).
- Balance Drawdown Maximal — difference in deposit currency between the highest local balance value and the next lowest account balance value. The maximal drawdown value in percentage is given in brackets. $\text{MaximumDrawDown} = \text{Max}[\text{Local High} - \text{Next Local Low}]$ See the [drawdown calculation example](#).
- Balance Drawdown Relative — difference in percentage terms between the highest local balance value and the next lowest account balance value. The maximal drawdown value in monetary terms is given in brackets. $\text{RelativeDrawdown} = \text{Max}[(\text{Local High} - \text{Next Local Low}) / \text{Local High} * 100]$ See the [drawdown calculation example](#).
- Equity Drawdown Absolute — difference between the initial deposit and the minimal level below initial deposit throughout the whole testing period. The calculation is similar to that of the Balance Drawdown Absolute.
- Equity Drawdown Maximal — difference in deposit currency between the highest local equity value and the next lowest equity value. The maximal drawdown value in percentage is given in brackets. The calculation is similar to that of the Balance Drawdown Maximal.
- Equity Drawdown Relative — difference in percentage terms between the highest local equity value and the next lowest equity value. The maximal drawdown value in monetary terms is given in brackets. The calculation is similar to that of the Balance Drawdown Relative.
- Profit Factor — ratio of the gross profit to the gross loss. A value of one means that these parameters are equal;
- Recovery Factor — the value reflects the riskiness of the strategy, i.e. the amount of money risked by the Expert Advisor to make the profit it obtained. It is calculated as the ratio of gained profit to the maximum drawdown;
- AHPR — arithmetic mean of a trade (change in percents). Arithmetic mean of equity changes per trade. The arithmetic mean usually overestimates the profitability of a trading system as compared to the geometric mean. If the geometric mean implies the multiplication of results of each trade, the arithmetic mean just sums them. The value in percents is given in brackets. It is positive if the trading system is profitable. The negative value means that the system is losing.
- GHPR — geometric mean of a trade (change in percents). Geometric mean shows by how many times the capital changed after each trade in average. The relative equity change is often a more objective

estimation than the expected payoff. Capital change in percents is given in brackets. A negative number in brackets means that on the average the capital is reduced on each trade.

- Expected Payoff — a statistically calculated value showing the average return of one deal. Also, it is considered to display the expected return of the next trade;
- Sharpe Ratio — a classic measure which is commonly used to evaluate the performance of a portfolio manager, fund results or a trading system. The ratio is calculated as $(\text{Return} - \text{Risk-Free Rate}) / \text{Standard Deviation of Return}$. In the strategy tester, the Risk-Free Rate is assumed to be zero. The ratio values are usually interpreted as follows:
- Sharpe Ratio < 0 — the strategy is unprofitable. Bad.
- $0 < \text{Sharpe Ratio} < 1.0$ — the risk does not pay off. Such strategies can be considered when there are no alternatives. Indefinite.
- Sharpe Ratio ≥ 1.0 — this can mean that the risk pays off and that the portfolio/strategy can show results. Good.
- Sharpe Ratio ≥ 3.0 — a high value indicates that the probability of obtaining a loss in each particular deal is very low. Very good.
- LR Correlation — linear regression correlation. A balance graph is a broken line, which can be approximated by a straight line. To find the coordinates of the straight line, the least-squares method is applied. The resulting straight line is called "linear regression" and allows estimating the deviation of balance graph points from the linear regression. Correlation between the balance graph and the linear regression allows to estimate the degree of the capital variability. The less sharp peaks and troughs on the balance curve, the closer the parameter value is to 1. Values close to zero mean the random nature of trading.
- LR Standard Error — the standard error of balance deviation from the linear regression. This index is used to estimate the balance chart deviation from the linear regression in money terms. It only makes sense to compare systems with similar initial conditions (the same values of the initial equity). The higher the value, the more balance deviates from a straight line.
- Margin Level — minimal level of margin in percentage terms registered during testing;
- Z-Score — series testing (the probability of correlation between trades). The series testing allows to estimate the degree of correlation between trades and evaluate whether the trade history includes more/less periods of consecutive profits/losses than normal distribution implies. The detected correlation allows to apply the methods of money management and/or change the trading system algorithm to maximize profit and/or to remove the dependence. Both non-finding the real correlation and finding a nonexistent correlation between trades are dangerous. The Z score indicates deviation from normal

distribution in the sigma. A value above 3 indicates that a win will be followed by a loss with the probability of 3 sigma (99.67%). A value below -3 indicates that a win will be followed by a win with the probability of 3 sigma (99.67%).

- OnTester Result — a value returned by the OnTester function in the Expert Advisor as a result of testing. It corresponds to the **custom criterion** of optimization;
- Total Trades — the total number of trades (deals resulted in fixing a profit/loss);
- (Total Deals) — the total number of deals;
- Short Trades (won %) — number of trades that resulted in profit from selling a financial instrument, and percentage of profitable short trades;
- Long Trades (won %) — number of trades that resulted in profit from purchasing a financial instrument, and percentage of profitable long trades;
- Profit Trades (% of total) — the amount of profitable trades and their percentage in the total trades;
- Loss Trades (% of total) — the amount of losing trades and their percentage in the total trades;
- Largest profit trade — the largest profit of all profitable trades;
- Largest loss trade — the largest loss of all loss-making trades;
- Average profit trade — the average profit value per a trade (the total of profits divided by the number of winning trades);
- Average loss trade — the average loss value per a trade (the total of losses divided by the number of losing trades);
- Maximum consecutive wins (\$) — the longest series of winning trades and their total profit;
- Maximum consecutive losses (\$) — the longest series of losing trades and their total loss;
- Maximal consecutive profit (count) — the maximum profit of a series of profitable trades and the amount of profitable trades in this series;
- Maximal consecutive loss (count) — the maximal loss of a series of losing trades and the number of losing trades in it;
- Average consecutive wins — the average number of winning trades in profitable series;
- Average consecutive losses — the average number of losing trades in losing series.
- Correlation (Profits, MFE) — correlation between returns and the MFE (Maximum Favorable Excursion, maximum size of a potential profit occurred during the life time of a position). Each position had its maximal profit and maximal loss between opening and closing. MFE shows profit in the favorable excursion of the price. Each position has its result and two parameters — MFE and MAE (Maximum Adverse Excursion, maximum size of a potential loss occurred during the life

time of a position). Thus, each position can be drawn on a plane where MFE is plotted along the X axis, the result is plotted along the Y-axis. Results close to MFE mean the most complete use of the favorable price excursion. A straight line on the graph shows approximation by function $\text{Profit} = A * \text{MFE} + B$. $\text{Correlation}(\text{Profits}, \text{MFE})$ allows to estimate relation between the profits/losses and the MFE. Values close to 1 mean that trades fit well into the approximation line. Values close to zero mean weak correlation. MFE characterizes the ability to realize potential profit.

- **Correlation (Profits, MAE)** — correlation between results and MAE (Maximum Adverse Excursion). Each position reached its maximal profit and maximal loss between opening and closing. MAE shows the loss during the adverse excursion of the price. Each position has its result and two parameters — MFE and MAE. Thus, each position can be drawn on a plane where MAE is plotted along the X axis, the return is plotted along the Y axis. Results close to MAE mean the most complete protection against adverse price excursion. A straight line on the graph shows approximation by function $\text{Profit} = A * \text{MAE} + B$. The $\text{Correlation}(\text{Profits}, \text{MAE})$ allows to estimate relation between the profits/losses and the MAE. Values close to 1 mean that trades fit well into the approximation line. Values close to zero mean weak correlation. MAE describes the drawdown during the position lifetime and best characterizes the use of protective Stop Loss.
- **Correlation (MFE, MAE)** — correlation between MFE and MAE. It shows correlation between two rows of characteristics. The ideal value is 1 - we take the maximum profit and protect the position throughout its lifetime. A value close to zero indicates there is practically no correlation.
- **Minimal position holding time** — a minimum amount of time between opening a position and closing it completely. Complete closing of a position is its full elimination; the calculated value does not take into account partial closing or position reversal.
- **Maximal position holding time** — a maximum amount of time between opening a position and closing it completely.
- **Average position holding time** — the average time between opening a position and closing it completely during testing.

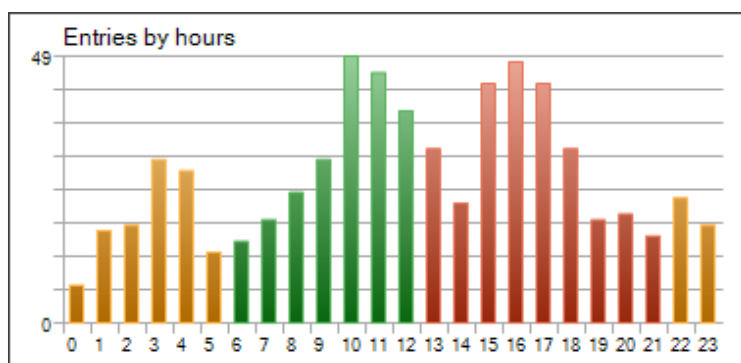
If withdrawal operations are performed in an Expert Advisor during testing/optimization, the **drawdown** rates are calculated taking into account these operations.

The drawdown values calculated before withdrawing are memorized by the program. During withdrawal, their calculation will be restarted on the basis of the current values of balance and equity. If new calculated drawdown values are greater than the ones saved before, the program will remember these new values. So the highest drawdown value is included into the final report.

Diagrams

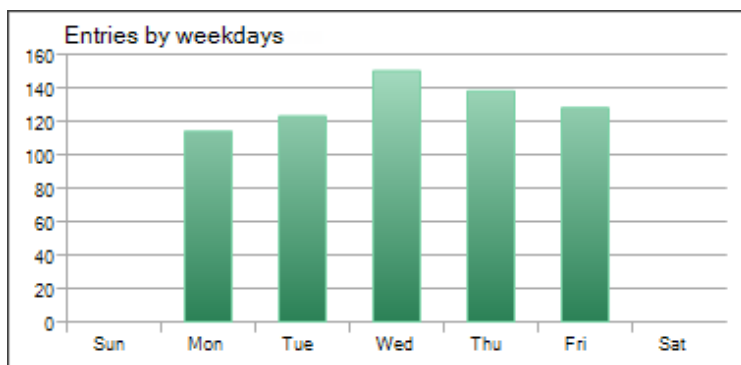
The following diagrams are available in the testing report:

Entries by hours



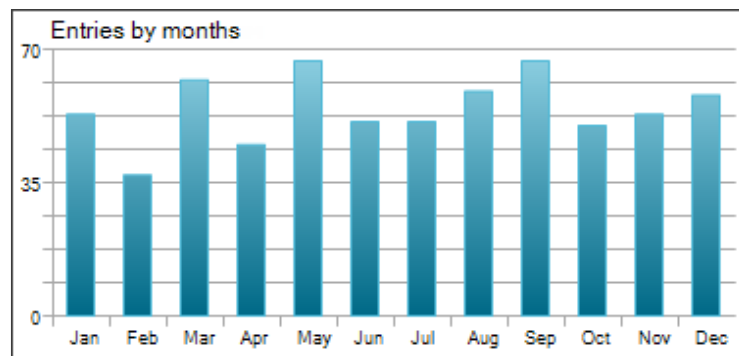
This diagram shows the distribution of market entry deals (opening, increase and reversal of positions) by hours. The colors of the diagram bars mark trading sessions: Asian (yellow), European (green) and American (red).

Entries by weekdays



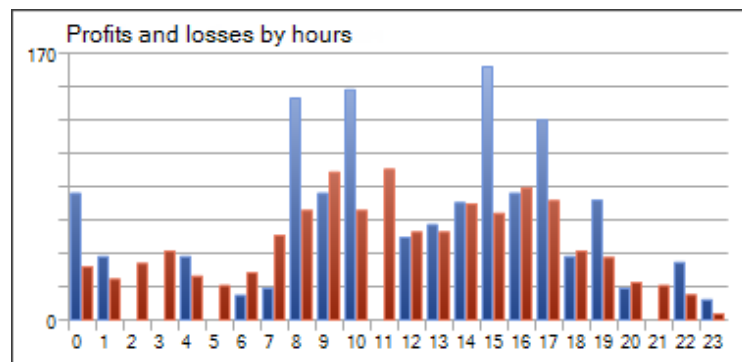
This diagram shows the distribution of market entry deals (opening, increase and reversal of positions) by days of the week.

Entries by month



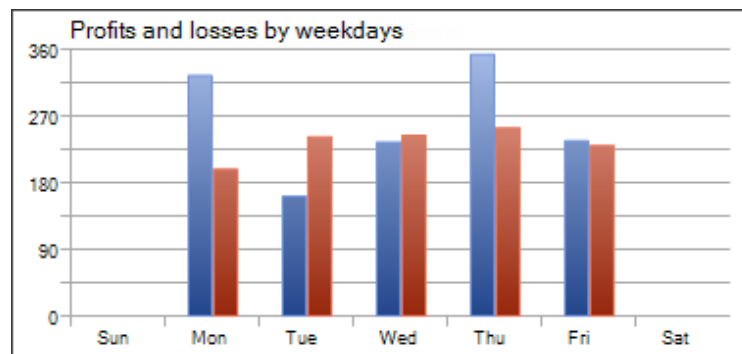
This diagram shows the distribution of market entry deals (opening, increase and reversal of positions) by months.

Profits and losses by hours



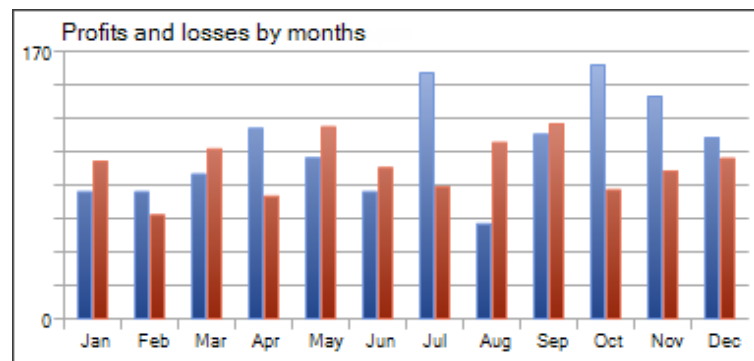
This diagram shows the distribution of market exit deals (closure, partial closure and reversal of positions) by hours. The colors of the diagram bars show profitable (blue) and losing (red) deals.

Profits and losses by weekdays



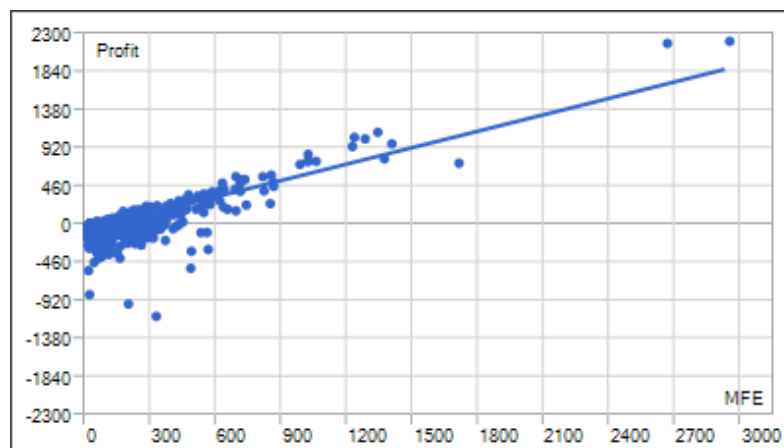
This diagram shows the distribution of market exit deals (closure, partial closure and reversal of positions) by weekdays. The colors of the diagram bars show profitable (blue) and losing (red) deals.

Profits and losses by months



This diagram shows the distribution of market exit deals (closure, partial closure and reversal of positions) by months. The colors of the diagram bars show profitable (blue) and losing (red) deals.

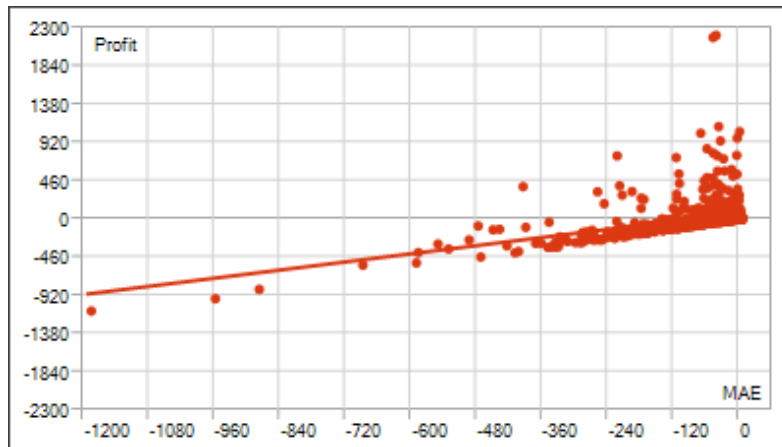
MFE-Profits Distribution



Positions are plotted as dots on the graph of MFE (Maximum Favorable Excursion) — Profits. Values of both axes are given in the deposit currency. In addition to the profit value of each position including swaps plotted along the Y axis, the graph shows the maximally possible profit during the position holding time. It allows to estimate the quality of protection of the paper (unrealized) profit.

Though the distribution of points along the graph provides a picture of the trading system, a linear regression, which is approximation by least squares, is given for an objective assessment. Ideally, the line should go at an angle of 45 degrees.

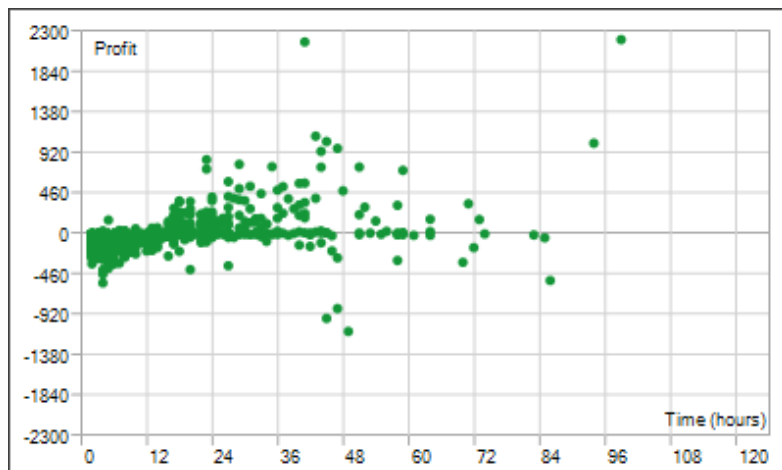
MAE-Profits Distribution



Positions are plotted as dots on the graph of MAE (Maximum Adverse Excursion) — Profit. Values of both axes are given in the deposit currency. In addition to the profit value of each position including swaps plotted along the Y axis, the graph shows the highest drawdown during the position holding time. It allows estimating trades in terms of drawdown outstaying.

Though the distribution of points along the graph provides a picture of the trading system, a linear regression, which is approximation by least squares, is given for an objective assessment. The less trades with negative X (MAE) values, the better. The graphical analysis helps to estimate the maximum tolerated loss, after which the possibility of taking profit is very small (if the analysis is performed for one currency pair and in points).

Profit and position holding time distribution



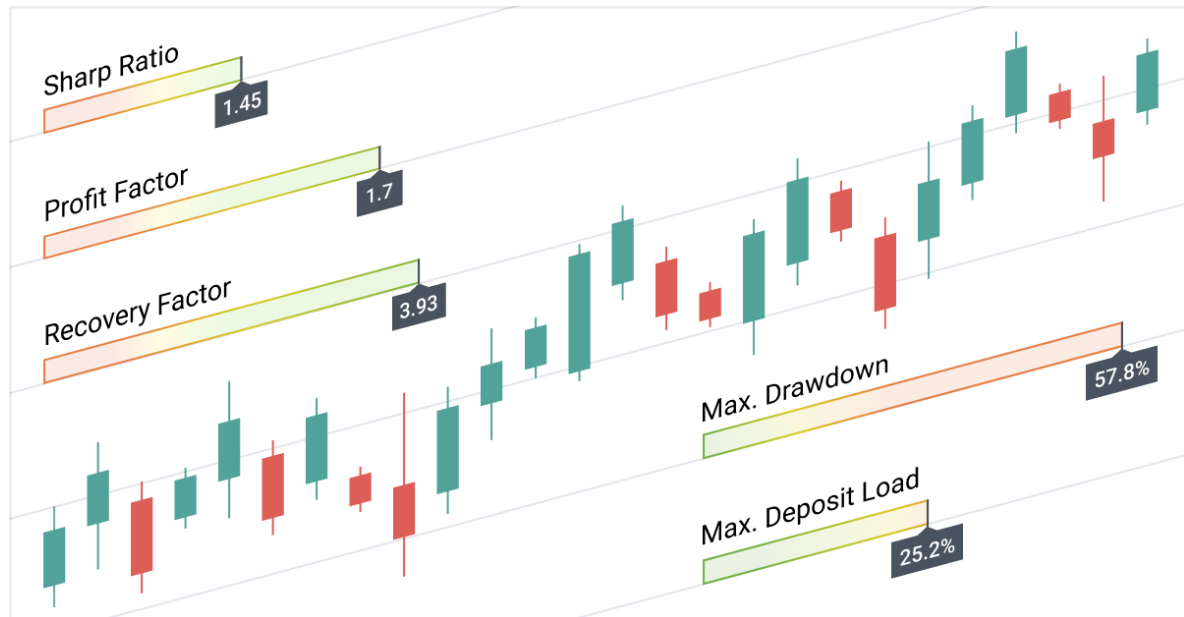
Points plotted on the Profit — Time graph indicate positions. The graph displays a correlation between the position holding time and the profit obtained as a result of closing it. Values on the time axis can be given in seconds, minutes or hours depending on the scale required. Profit is displayed in the deposit currency. The position holding time is calculated as the time from its opening till complete closing. Complete closing of a position is its full elimination; the calculated value does not take into account partial closing or position reversal.

New MetaTrader Report: 5 key trading metrics

In this article, we will look at five key performance indices which characterize trading efficiency and stability

6 October 2023

Successful trading in financial markets requires thoughtful and well-founded decision-making. Comprehensive performance analytics can assist traders in making informed investment decisions and in optimizing their strategies. In this article, we will look at five key performance indices which characterize trading efficiency and stability.



1. Sharpe Ratio

- What it is: A measure of strategy profitability in relation to risk.
- Why it is important: A high Sharpe Ratio value indicates that a strategy can efficiently generate profits with minimal risks. The higher the ratio, the better.
- How to interpret: A value greater than 1 indicates that the strategy's returns justify the associated risks. Ratio values of 3 or higher indicate an excellent strategy.

Below is a successful trading report which shows an annual growth of more than 1,500%. Balance and growth diagrams are quite smooth, without sharp drops and spikes. However, the strategy cannot be considered stable since the Sharpe Ratio is 0.21, which means that the trader is taking very high risks to achieve these trading results.

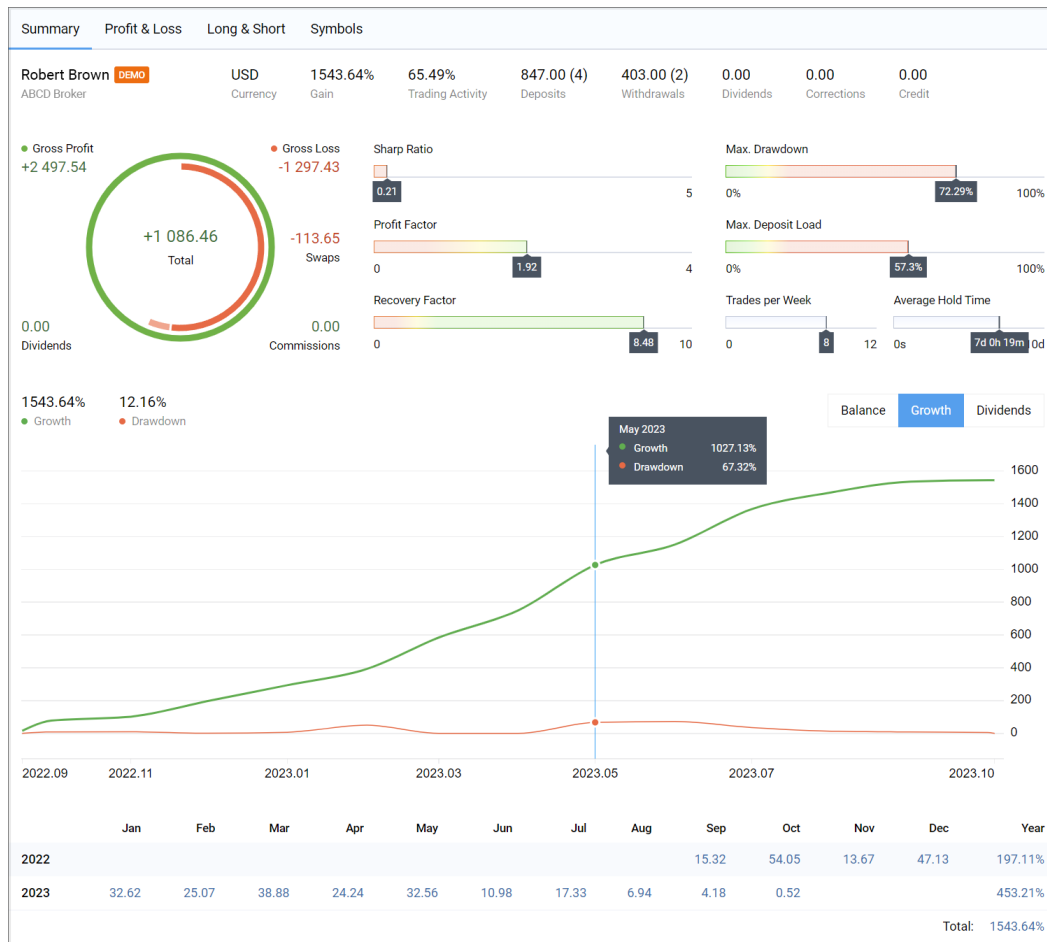
The Sharpe ratio is interpreted as follows:

- Sharpe Ratio < 0 Poor. The strategy is generally unprofitable.
- Sharpe Ratio < 1.0 Undefined. The risks are not justified. The use of such strategies can only be considered when there are no alternatives.
- Sharpe Ratio ≥ 1.0 Good. If the Sharpe Ratio is higher than one, it means that the risks are justified by the portfolio/strategy

performance.

- Sharpe Ratio ≥ 3.0 Excellent. A high ratio indicates lower probability of losses in each individual trade.

If the Sharpe ratio is less than zero, the strategy is unprofitable. However, even without analyzing the Sharpe Ratio, such poor results can be evident from balance and growth diagrams only.



2. Maximum Drawdown

- What it is: The maximum decline in balance from its highest value.
- Why it is important: The metric shows the portion of capital at risk during the worst trading period.
- How to interpret: Lower drawdown values indicate a more reliable strategy.

The maximum drawdown in the above report is equal to 72.3%, which indicates a very high risk. The diagram above shows a drawdown of 67% in May 2023. This risk value is considered extreme since the optimal drawdown value should not exceed 20-30%.

Why should the optimal drawdown be no more than 30% (preferably less than 20%)? When choosing a trading strategy, you should assume that the historically registered drawdown can actually be surpassed by a factor of 2 or more. If the maximum drawdown is 30%, the future value can reach 60% ($2 \times 30\%$). A trading account can endure such a drawdown value. However, if the maximum drawdown is 72%, then it may well be 100% in the future, which means that all the funds could be lost.

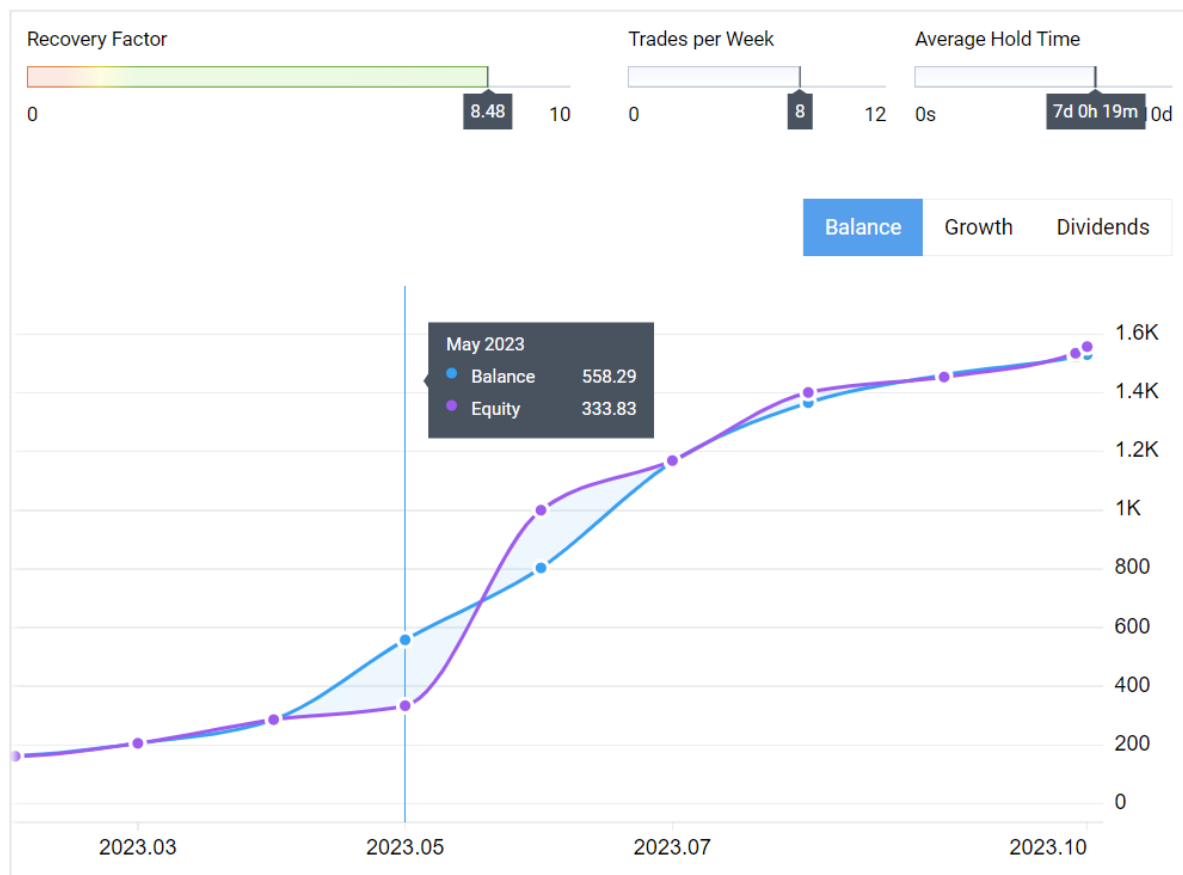
3. Recovery Factor

- What it is: The ratio of generated profit to maximum drawdown.
- Why it is important: The factor shows the ability of the strategy to recover from losses.
- How to interpret: The higher the Recovery Factor, the faster the strategy covers losses and begins to generate profits.

The Recovery Factor in the above report is equal to 8.26 and is in the green zone, which is a good value. Ideally, a trading strategy should have a Recovery Factor greater than 3, with greater values being more desirable.

The Summary report in the funds mode shows that the account equity (USD 333.83) is significantly lower than the balance (USD 558.29). The account had a large drawdown, but the resulting profit provided a good recovery factor.

A strategy with a high Recovery Factor (3 or more) indicates that the current drawdown can be compensated by future profits. If the Recovery Factor is less than 1, then several consecutive drawdowns may potentially cause the account balance to drop to zero.



4. Profit Factor

- What it is: The ratio of total profits to total losses.
- Why it is important: The factor shows the potential long-term efficiency of the strategy.
- How to interpret: Values greater than 1 indicate profitability.

The primary purpose of financial trading is to generate profits. While any trading strategy may have winning and losing trades, profits from winning trades should exceed losses from unprofitable ones. The Profit Factor precisely indicates how much the total generated profits exceed the total losses.

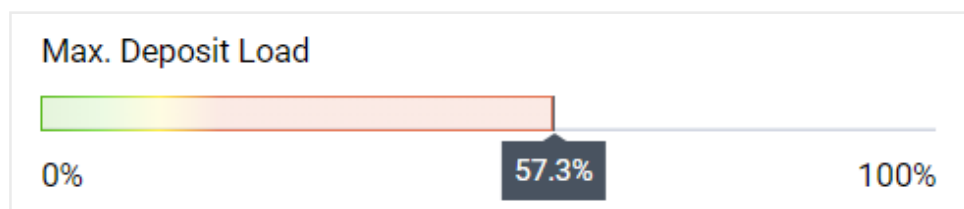
The Profit Factor in this report is equal to 1.91, which is a good value. Accordingly, the diagram is in the green zone.



5. Max. Deposit Load

- What it is: The highest percentage of account funds utilized for open positions.
- Why it is important: High deposit loads carry additional risks for a trading account. If the market moves in an unfavorable direction, the probability of losing all funds increases if the deposit loads are high.
- How to interpret: Lower values indicate reduced exposure to potential losses.

There is no optimal deposit load value which is suitable for all strategies. The load depends on the specific trading style and position holding times. Scalping strategies can utilize higher deposit shares since they aim at generating profits from small price changes and usually open increased-volume positions. For margin trading, including Forex, the margin requirement is significantly lower due to the provided leverage. This allows for increased profits from small price changes but at the same time carries higher risks of potential losses if the currency pair moves in an unfavorable direction. For example, if the leverage is 100, then the margin required to open a position is reduced by 100 times. Consequently, if a trader opens a position for the entire deposit and the price moves 100 points in the opposite direction, the entire deposit will be lost. Furthermore, many brokers set a Stop Out level of 50%, which means that 50 points in an unfavorable direction will lead to large losses. Such a movement is typically within the daily range for most currency pairs. The maximum deposit load in this report is 57%. Such trading is risky and thus the metric is in the red zone.



Summary

The Summary page features key indices, including balance/equity graphs and growth percentage dynamics. Analysis is summarized as below:

- Growth and balance graphs are quite smooth, without sharp fluctuations — ✓
- Sharpe Ratio = 0.21 — X
- Maximum Drawdown = 72% — X

- Recovery Factor = 8.26 — ✓
- Profit Factor = 1.91 — ✓
- Max. Deposit Load = 57% — ✓

While three out of the five metrics are promising, the trading strategy cannot be considered stable and safe. The generated profits are associated with high risks.

All of the considered statistics are important for the analysis and evaluation of trading strategies. You should always analyze a combination of performance indices, taking into account the specific trading strategy features, and market conditions. Proper use and understanding of statistical metrics can assist in minimizing risks and in improving trading results.

With the updated MetaTrader 5 trading report, traders can evaluate their strategy performance directly in the terminal. Such comprehensive analytics can highlight strategies' positive aspects, and prompt at weak areas, which if improved can reduce risks and enhance trading stability.